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Research Statement

LLM Fine-Tuning for Mechanics

My interest in scientific machine learning began with a language model fine-tuning project focused on answering mechanics of materials questions. I studied the organization of existing instruction datasets and prepared question-answer pairs and concept explanations based on textbooks and technical literature. Using these data, I fine-tuned Qwen2.5-7B with LoRA and released two instruction datasets and two model checkpoints on Hugging Face. Through preparing these training samples, I learned the importance of connecting learning objectives with data design. This experience motivated my subsequent work on learning-based methods for predicting physical fields.

ConvLSTM Prediction of Battery Material Fields

In a project on battery materials, I automated MATLAB–COMSOL simulations to generate concentration and stress field sequences. I trained separate conditional ConvLSTM models in PyTorch to predict these fields and compared multistep predictions with simulation results to examine autoregressive error accumulation. To make training closer to actual prediction conditions, I used scheduled sampling to gradually introduce the model’s own predictions as inputs, allowing the model to encounter the type of inputs it would receive during autoregressive prediction. This project led me to consider how training strategies should account for the conditions under which a model will be used.

PINNs for Shock Capturing

I later explored how physical knowledge could guide learning-based models through a physics-informed neural network (PINN) for shock-capturing problems governed by the one-dimensional Burgers equation. I implemented the PINN in PyTorch and used automatic differentiation to compute the derivatives required for the governing equation residual, incorporating the physical constraints into the training process. Using the analytical inviscid solution as a reference, I compared standard PINNs with PINNs using fixed artificial viscosity. Although artificial viscosity reduced the relative L^2 error, it increased the maximum pointwise error, revealing that improved global accuracy did not necessarily correspond to better local predictions. This experience made me more careful about evaluating scientific machine learning models, where numerical accuracy, local behavior, and physical assumptions all need to be considered together.

Hybrid FEM–Neural Operator Solvers

I am currently developing a hybrid finite element–neural operator solver based on non-overlapping domain decomposition. I use GPU-accelerated JAX-FEM simulations to generate training data and train Transolver in JAX to predict subdomain stresses from interface displacements. The predicted stresses are then used to compute interface forces, which are exchanged with the finite element solver. The updated finite element solution provides new interface displacements for the next neural operator prediction. Through this feedback process, the neural operator becomes part of an iterative physical solver rather than an independent prediction model. Therefore, evaluating only prediction accuracy on standalone samples is insufficient to determine its reliability in a coupled simulation.

Because the neural operator receives changing interface conditions during coupling iterations, I began to consider whether the training data could represent the inputs encountered during the actual solution process. This consideration guided my design of Gaussian random field samples, aiming to better capture the range of interface displacements provided to the neural operator. I also investigated convergence acceleration strategies and relaxation parameters, since the stability of the coupled solver depends not only on prediction accuracy but also on how information is exchanged between the neural operator and the finite element solver. This work changed my perspective on scientific machine learning: developing a useful model requires considering the training data, the computational process, and the final application together.

Working with JAX-FEM and JAX deepened my understanding of differentiable computation. When the relevant operations are differentiable, automatic differentiation propagates gradients through the computational graph using the chain rule. A loss measures the difference between predicted and observed responses; its gradients with respect to unknown material parameters or parameterized boundary loads can guide their updates. This extended my interest from forward simulation to parameter identification.

STEM Outreach

As part of a national-level student innovation project, I helped peers learn language-model fine-tuning, drawing on my work in mechanics of materials. My public repositories and website document the ConvLSTM simulation-to-prediction workflow and provide PINN code with analytical comparisons. I aim to explain methods and their reasoning clearly so others can understand and use them. As I continue learning and conducting research, I hope to share my experience in mechanics, model training, and research with peers, work through problems together, and improve my research judgment through discussion.

Future Work

I am also interested in differentiable computation that connects physical simulations with learning models, particularly how gradients can be propagated through computational graphs to enable optimization and learning. As I continue my research, I hope to deepen my understanding of mechanics and numerical methods and gradually develop into an independent researcher. I want to learn how to formulate meaningful research questions, critically examine existing methods, and design and carefully evaluate new computational approaches. In the long term, I hope to pursue research and development in companies such as NVIDIA, where I can help translate advances in scientific machine learning and accelerated simulation into practical computational tools.